

TMA4267 LinStat DOE Project

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1 Introduction

In this project, we will plan, perform and analyse a statistical experiment. We will use Design Of Experiment with 3 factors and 2 levels where the response is continuous so that the experiment can be analysed with linear regression.

The goal of this experiment is to determine how different factors influence the probability of Type 1 error, that is, falsely rejecting the null hypothesis H_0 . In research, statistical hypothesis testing is widely used. However, but the reliability of tests depends on factors such as sample size, data distribution, and the choice of test statistic.

2 Our Experiment

As mentioned above, the aim of our project is to investigate how false rejection of H_0 is affected by 3 factors with 2 levels each. In our simulation we vary these factors using a 2^3 factorial design and assess their main affect as well as their interactions on the estimated error rates.

Initially, we planned to have only two replicates per factor combination, but the this led to inconsistent findings. This is discussed in more detail later in the report. Therefore to improve statistical reliability, we will conduct 10 replicates per condition and repeat each hypothesis test 100 times per run. In other words, we will have 100 simulations and 80 observations. This is to our advantage as more observations will give us more deegrees of freedom (particulary $80 - 8 = 73$ degrees of freedom) to explain variation in the model. Moreover, having more degrees of freedom means that standard errors of the effect estimates are smaller, making it easier to detect significant effects.

2.1 Selection of factors and levels

The three factors we chose for this experiment are sample size, distribution, and test type. The factors and their leveles are summarized in the table below.

Factor	Low (-1)	High (+1)
A: Sample Size	5 (Small)	100 (Large)
B: Distribution	Normal ($\mathcal{N}(0, 1)$)	t -distribution (df=3)
C: Test Type	t -test	Z -test

In this experiment small sample sizes are expected to increase variability and potentially increase the Type 1 Error rate, while larger sample sizes provide more stable estimate. The normal distribution is used as a baseline for the tests we will use, but using the t -distribution with 3 degrees of freedom induces heavy tails, which challenge the assumption on normality and thus can affect the type 1 error rates. Furthermore, while the t -test is typically used for small sample sizes with unknown variance, the Z -test(assuming known variance) is more appropriate on larger samples.

2.2 Hypothesis and Interactions

The main hypothesis is that the violation of the normality assumption and the sample size will influence the type 1 Error rate differently for the two testing procedures. In particular:

Factor A and factor C interaction: The Z-test assumes normality, thus when the data comes from the t-distribution the error rate is expected to deviate more significantly from the normal level.

Factor A and factor B interaction: With small sample sizes the Central Limit Theorem (CLT) may not sufficiently normalize the data drawn from the t-distribution which can amplify differences in error control for the two tests. For larger samples CLT might reduce this discrepancy.

Factor B and factor C interaction: Both the z-test and the t-test assume normality from the underlying data. Drawing from a t-distribution violates this assumption. In this case, the Z-test which relies more heavily on the normality assumption and use of known variance is expected to be more affected than the t-test.

2.3 Model and design

Our response variable y represents the Type 1 Error rate (probability of falsely rejecting H_0) and is modeled as

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_{12} x_1 x_2 + \beta_{13} x_1 x_3 + \beta_{23} x_2 x_3 + \beta_{123} x_1 x_2 x_3 + \varepsilon, \varepsilon \sim N(0, \sigma^2)$$

Here x_1, x_2, x_3 corresponds to the factors A(sample size), B(Distribution), C(Test type). The interaction terms (eg. $x_1 x_2$) capture the dependency between factors.

Design implementation details: In the initial idea for this experiment we had a 2^3 factorial design, with 16 observations where each combination was replicated twice. However, during analysis of our data we found that this only yielded significant results about 50% of the time. Therefore, we now employ 10 replicates per experimental condition. These are randomized in order before performing each run of the simulation. Moreover, for each experimental run the hypothesis test is repeated 100 times to estimate the type 1 error rate with greater prediction.

Simulation and analysis loop: The complete experiment is simulated over 100 independent runs. For each run, a linear model is fitted to the data and key statistics are recorded, this is to be able to analyse the overall behaviour of the model and experimental results.

2.4 Blocking

Blocking is not necessary in our experiment because we do not need to split the simulation into two separate groups. Blocking is typically used when there is a factor that could introduce variability. Then we would require the experiment to be split into different blocks to control for it. In our experiment however, the entire experiment is simulated in one go, which means there is no need to split the data into blocks. All factors are applied consistently across all runs. Additionally, we do not have external factors that could introduce variation.

2.5 Hypotheses in The Experiment

The following hypotheses are tested in the experiment:

H_0 : The two samples come from the same distribution

H_1 : The two samples come from different distributions

Our aim is to analyze type 1 error and in order to estimate the Type 1 error rate, we must always generate data where H_0 is true. Type 1 error is when we incorrectly reject the null hypothesis, H_0 , when it is actually true. Since H_0 is true, any rejection is a Type 1 error.

We will count how often the p-value is below 0.05, meaning we reject H_0 when we should not have.

In this hypothesis test, we will use the following test-statistics of the Z-test and t-test for two samples defined as

$$Z = \frac{X_1 - X_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}}$$

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where we have that:

$\bar{X}_1, \bar{X}_2$ are the sample means,
 s_1^2, s_2^2 the sample variances,
 n_1, n_2 the sample sizes and finally
 σ_1^2 and σ_2^2 are the variances of the two samples.

2.6 Analysis of data

In our simulation we have gathered diagnostics on the data and model. In figure 1 we see that the p-value of the model is well below the threshold of 0.05 for all runs of the simulation. This means that the model is regarded statistically significant in all cases. Further, in figure 3 we see that the intercept, the influence of A, C and the interaction between A and C are significant in all runs, whereas the other coefficient values are significant in less than 10% of the runs. This means that the value of these interactions can not be regarded as having a large effect on the results. In figure 2 we see the QQ plots for the first 9 simulations. From this we can see that the residuals in general follow the red reference line. There seems to be some tail effects on the upper and lower tail, which could mean a slight deviation from the normal distribution of the residuals. We also see that the plots are consistent across runs, with only minor differences in how the points cluster around the lines-which are expected due to the stochastic nature of the experiments. Thus the normality assumption for the error term in the linear model is not severely violated. This means that the p-values obtained for the model are credible, with the reservation that outliers and heavy tails may occur-as we have seen in the discussion on the significance of the different coefficient values.

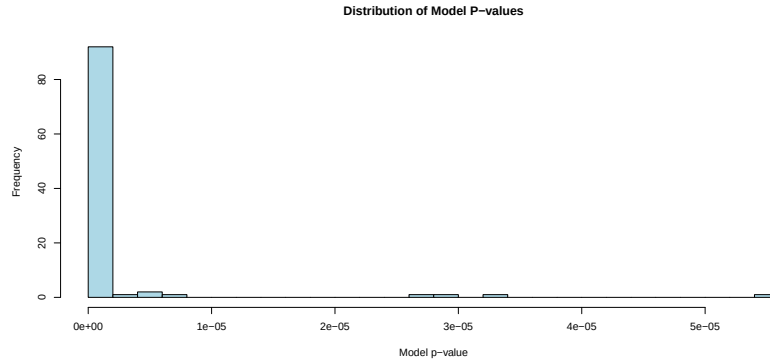


Figure 1: Distribution of the p-values for the model across 100 simulations

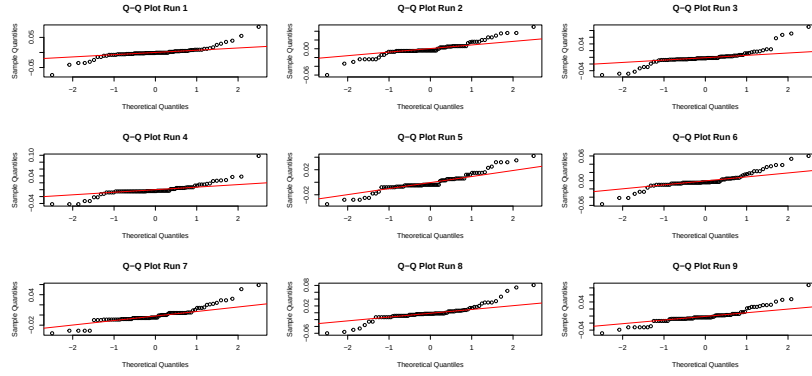


Figure 2: QQ plots for the first 9 simulations

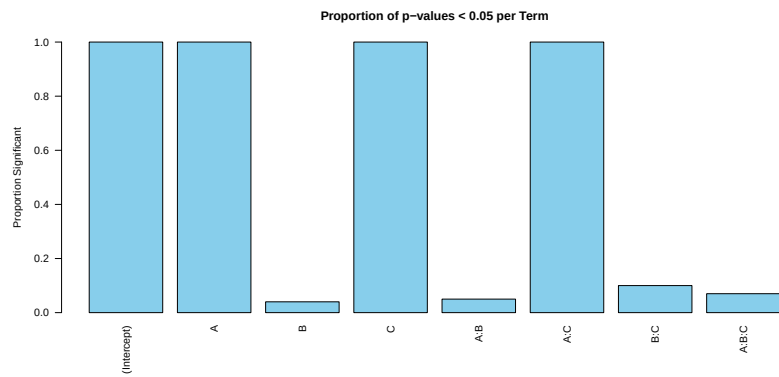


Figure 3: Proportion of time the coefficient estimates are statistically significant

3 Results

In this section, we will present the results obtained for the effect of the different factors, and the interactions between them. In figures 5 and 4 we can clearly see that the values of A, C and the interaction between them have the largest average value and effect on the rejection of the null hypothesis. This is backed up by figure ??, where we can see that changing from a small to a larger sample size has a negative effect on the rejection rate. Similarly we see from the same plots that changing factor C from t-test to Z-test has a positive influence on the rejection of the null-hypothesis. Further, we see in figure 4 that the values of the other coefficients are centered around 0 with smaller distributions. We also get that the intercept, representing the mean of Type 1 error rates have a value of aproximatly 0.017 across experiments. From the interaction plot 7 we see a strong interaction between A and C, and barely any interaction for the other coefficient. Also here, the role of sample size A is visible.

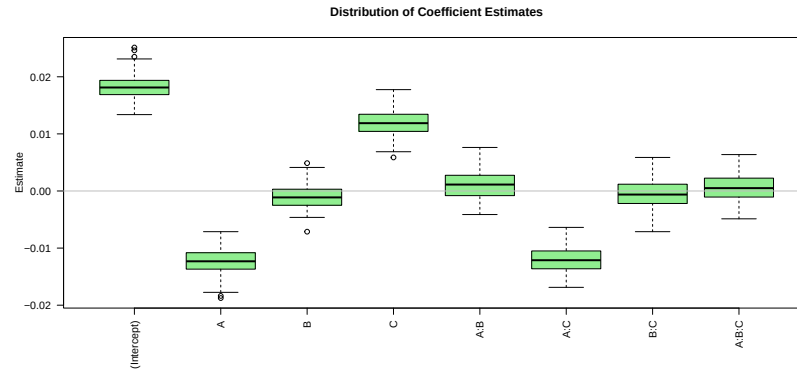


Figure 4: In this plot we can see how the Coefficient estimates are distributed through the 100 runs

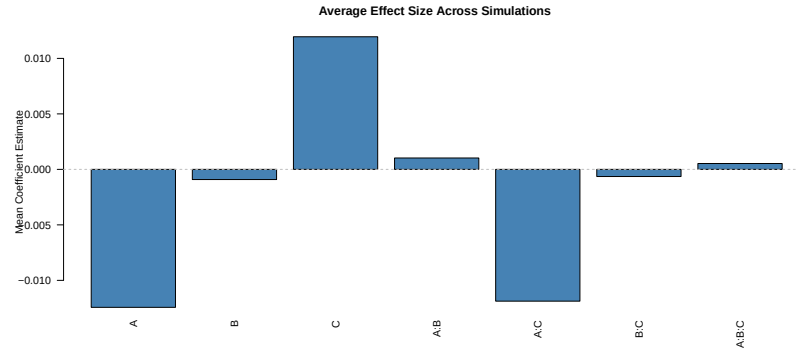


Figure 5: This plot shows the average values for the coefficent across runs

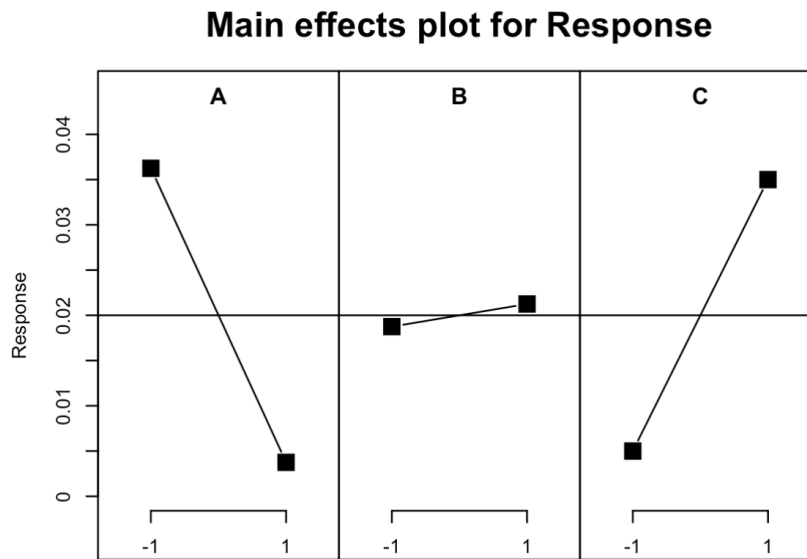


Figure 6: Main effect plot, single run

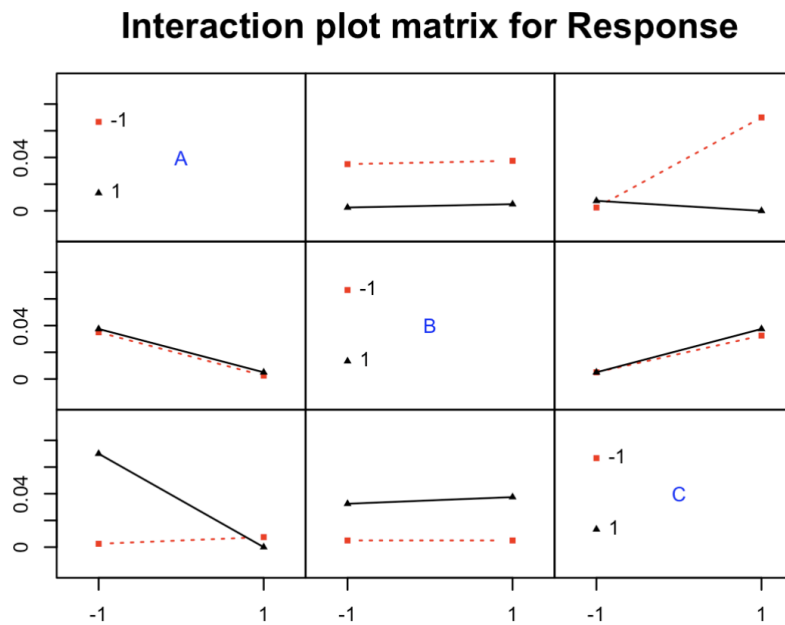


Figure 7: Interaction plot, single run

Below is the revised version, now split into a "Discussion" and a "Conclusion" section, with all original content preserved:

4 Discussion

One of the most surprising findings from this test is that Factor B (distribution type) does not have a significant impact on the rate of Type 1 Errors. The likely reason for this is that Type 1 Error is primarily controlled by the choice of test and the significance level α .

Furthermore, the impact of sample size and test type on the Type 1 Error rate is in line with our hypotheses. The interaction between these two is also not surprising, as the t -test and Z -test work best with small and large sample sizes, respectively. Larger sample sizes also mitigate errors due to drawing from a t -distribution, as the Central Limit Theorem (CLT) approximates normality in that case.

We also observed a small average interaction between Factor A (sample size) and Factor B (distribution), but this interaction is not statistically significant most of the time. This is surprising since our hypothesis stated the opposite. A possible explanation is that the interaction might be harder to distinguish in a standard factorial analysis, or that hidden confounding variables influenced our results. Moreover, the t -test and Z -test are known to be quite robust, and it appears that they can handle the t -distribution—even with a small sample size and 3 degrees of freedom.

5 Conclusion

In this project, we designed and analyzed a statistical experiment with three factors and two levels to investigate their influence on Type 1 error rates. The results we obtained provide insights into how factors such as sample size, data distribution, and test statistic affect the reliability of hypothesis testing.

Overall, our hypothesis aligns well with our statistical results. The experiment confirms that sample size and test type significantly influence the Type 1 error rate, while the data distribution (Factor B) appears to have a negligible effect. This suggests that the reliability of hypothesis testing is mainly determined by the choice of test and the significance level, with the CLT further mitigating potential issues when larger samples are used. Future research may explore refined experimental designs to better capture subtle interactions between sample size and distribution type.

6 Appendix: R Code for simulation and analysis

```
install.packages("FrF2")
library(FrF2)

# Type 1 error experiment

# Factor A = Sample size
fA <- function(levelA, n){
  if(levelA == -1){ # Low level
    data <- rnorm(n=5, mean = 0, sd=1)
  }
  else if(levelA == 1){ #High level
    data <- rnorm(n=300, mean= 0, sd=1)
  }
  return(data)
}

# Factor B = Distribution
fB <- function(levelB, data){
  if(levelB == -1){ # Low level, draw for a normal distribution
    transformed_data <- data # Data = normal distribution
  }
  else if(levelB == 1){
    transformed_data <- rt(length(data), df=3) #draw from a t-dist.
  }
  return (transformed_data)
}

# Factor C = Test type
fC <- function(levelC, data){
  p_values <- c()
  for(i in 1:100){
    # Randomly split data into two equal-sized samples, used in testing
    sample1 <- sample(data, length(data) / 2)
    sample2 <- sample(data, length(data) / 2)

    # But isnt sample only from normal distribution then? How do I get transformed_data
    # take t-distribution into account?

    if(levelC == -1){ # Low level, perform a t-test
      test <- t.test(sample1, sample2)
    }
    else if(levelC == +1){ # High level, perform z-test (assuming known variance)
      term_sqrt = var(sample1)/length(sample1) + var(sample2)/length(sample2)
      test_statistic <- (mean(sample1) - mean(sample2)) / sqrt(term_sqrt)
      p_value <- 2 * (1 - pnorm(abs(test_statistic)))
      test <- list(p.value = p_value)
    }

    p_values[i] <- test$p.value
  }

  # Note that fA generate raw data, fB transform the data depending on factor B, and
  # fC performs a hypothesis test on the transformed data
}
```



```

# Type I error rate
error_rate <- sum(p-values < 0.05) / 100
return(error_rate)
}

# 2^3 Factorial experiment
experiment <- function(n, r){ # n is sample size, r is row of factor levels
  # r can for example look like: r = [1, -1, 1]
  # r is a row from X123
  data <- fA(r[1], n)
  transformed_data <- fB(r[2], data)
  error_rate <- fC(r[3], transformed_data)

  return(error_rate)
}

# Design matrix
X <- cbind(
  c(-1, -1, -1, -1, +1, +1, +1, +1), # A
  c(-1, -1, +1, +1, -1, -1, +1, +1), # B
  c(-1, +1, -1, +1, -1, +1, -1, +1)  # C
)
colnames(X) <- c("A", "B", "C")

num_replicates <- 10
X_full <- X
for (i in 2:num_replicates) {
  X_full <- rbind(X_full, X)
}

colnames(X_full) <- c("A", "B", "C")

# One run for plots
random_order <- sample(1:nrow(X_full))
result <- c()

for(i in random_order){
  data <- fA(X_full[i, "A"])
  transformed_data <- fB(X_full[i, "B"], data)
  type1error <- fC(X_full[i, "C"], transformed_data)
  result <- c(result, type1error)
}
dataset <- data.frame(X_full[random_order, ], Response = result)

# Fit model
model <- lm(Response ~ A * B * C, data = dataset)
summary(model)
DanielPlot(model)
MEPlot(model)
IAPlot(model)

# — SIMULATION LOOP —
#Run the whole simulation 100 times to analyse overall behaviour
num_runs <- 100
coef_list <- list()
pval_list <- list()

```

```

model_pvals <- numeric(num_runs)
rsq_vec <- numeric(num_runs)
adj_rsq_vec <- numeric(num_runs)
resid_se_vec <- numeric(num_runs)
residual_list <- list()

for (i in 1:num_runs) {
  result <- c()
  random_order <- sample(1:nrow(X_full))

  for (j in random_order) {
    data <- fA(X_full[j, "A"])
    transformed_data <- fB(X_full[j, "B"], data)
    typeerror <- fC(X_full[j, "C"], transformed_data)
    result <- c(result, typeerror)
  }

  dataset <- data.frame(X_full[random_order, ], Response = result)
  model <- lm(Response ~ A * B * C, data = dataset)
  summary_model <- summary(model)

  coef_list[[i]] <- coef(summary_model)[, "Estimate"]
  pval_list[[i]] <- coef(summary_model)[, "Pr(>|t|)"]
  rsq_vec[i] <- summary_model$r.squared
  adj_rsq_vec[i] <- summary_model$adj.r.squared
  resid_se_vec[i] <- summary_model$sigma
  residual_list[[i]] <- residuals(model)

  f_stat <- summary_model$fstatistic
  p_model <- pf(f_stat[1], f_stat[2], f_stat[3], lower.tail = FALSE)
  model_pvals[i] <- p_model
}

# — ANALYSIS —

coef_df <- do.call(rbind, coef_list)
pval_df <- do.call(rbind, pval_list)

coef_stats <- apply(coef_df, 2, function(x) c(mean = mean(x), sd = sd(x)))
sig_props <- apply(pval_df, 2, function(p) mean(p < 0.05))

summary_table <- data.frame(
  Term = colnames(coef_df),
  Mean_Coefficient = coef_stats["mean", ],
  SD_Coefficient = coef_stats["sd", ],
  Prop_Significant_p_lt_0.05 = sig_props
)

print(summary_table)

# — PLOTS —
barplot(summary_table$Prop_Significant_p_lt_0.05,
  names.arg = summary_table$Term,
  las = 2,
  col = "skyblue",
  ylim = c(0, 1),

```

```

    main = "Proportion of p-values < 0.05 per Term",
    ylab = "Proportion Significant")

boxplot(coef_df,
        main = "Distribution of Coefficient Estimates",
        ylab = "Estimate",
        las = 2,
        col = "lightgreen")
abline(h = 0, col = "gray")

hist(model_pvals,
     main = "Distribution of Model P-values",
     xlab = "Model p-value",
     col = "lightblue",
     breaks = 20)
abline(v = 0.05, col = "red", lty = 2)

summary_table_no_intercept <- subset(summary_table, Term != "(Intercept)")

barplot(summary_table_no_intercept$Mean_Coefficient,
        names.arg = summary_table_no_intercept$Term,
        las = 2,
        col = "steelblue",
        main = "Average Effect Size Across Simulations",
        ylab = "Mean Coefficient Estimate")
abline(h = 0, col = "gray", lty = 2)

# — QQ PLOTS —
par(mfrow = c(3, 3))
for (i in 1:9) {
  qqnorm(residual_list[[i]], main = paste("Q-Q Plot Run", i))
  qqline(residual_list[[i]], col = "red")
}
par(mfrow = c(1, 1))

```